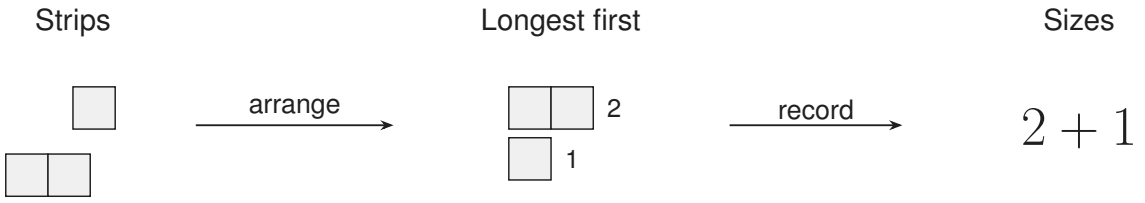
A collection uses all the units, in straight strips. Every strip has at least one unit. Moving or reordering strips does not make a new collection.



Problem 1: Find every collection you can make with 4 units, and with 5 units. Record each once, as rows or sizes.

4 units

5 units

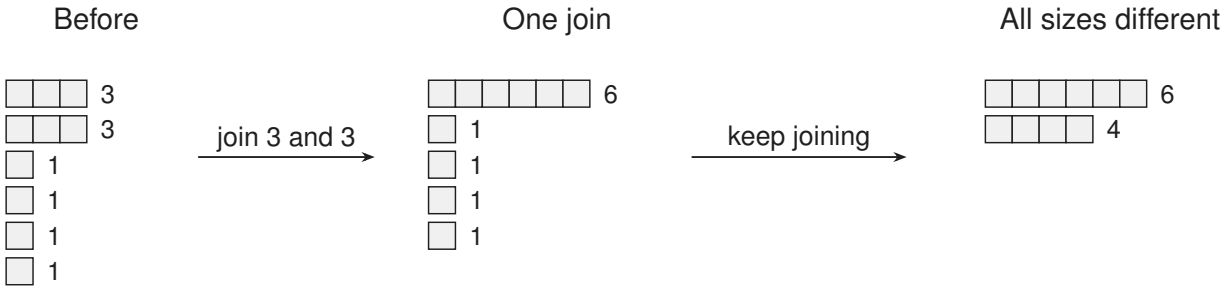
Odd sizes are $1, 3, 5, 7, \ldots$. "All sizes different" means no size repeats.

Problem 2: Choose 6 or 7 units. Find every collection of each kind below.

Total: _____ units

Only odd sizes	All sizes different

A join replaces two equal-size strips with one strip.



Problem 3: Build each collection. Join equal-size pairs until all sizes are different. Record the final collection.

Before	All sizes different
1, 1, 1, 1, 1, 1, 1, 1, 1	
5, 5, 3, 3, 3, 1, 1	
5, 3, 1	